



Guideline

Hypomagnesaemia

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical, Nursing, Pharmacy

High Risk Medicine: IV Magnesium is an injectable electrolyte

Introduction

This guideline provides hospital specific guidance to all Sir Charles Gairdner Osborne Park Hospital Care Group (SCGOPHCG) staff involved in the prescription, administration, supply and management of magnesium formulations in patients with hypomagnesaemia. This document supports and complies with related legislation, policy and other related documents [here](#).

1.0 General Information¹⁻³

Magnesium is a cation primarily found intracellularly within bone and is essential for membrane stability. Only around 1% of total body magnesium is found in extracellular fluid.

The PathWest reference range for plasma or serum magnesium is **0.70 to 1.10mmol/L**.

It should be noted that normomagnesaemic magnesium depletion (isolated cellular magnesium depletion) can cause refractory hypokalaemia or unexplained hypocalcaemia.

2.0 Clinical Presentation²

Mild magnesium deficiency is asymptomatic. Signs and symptoms of moderate to severe hypomagnesaemia are non-specific, but may include:

- cardiac arrhythmias
- convulsions
- muscle weakness
- neuropsychiatric changes
- tetany

It is essential to identify and minimise any underlying cause(s) when treating hypomagnesaemia.

Common Causes and Precipitating Factors	
Severe malnutrition	poor oral intake
Gastrointestinal losses	- diarrhoea - extensive bowel resection - intestinal fistulae - malabsorption - nasogastric suction - primary intestinal hypomagnesaemia - acute pancreatitis
Kidney losses	- chronic parenteral fluid therapy - hypercalcaemia and hypercalciuria - osmotic diuresis - post-kidney transplant
Drugs	- alcohol - aminoglycosides - amphotericin B - antibiotics - cisplatin - cyclosporin - diuretics - foscarnet - pentamidine - proton pump inhibitors
Other	- diuretic phase of acute kidney failure - Gitelman syndrome 'hungry bone' syndrome - phosphate depletion - post-obstructive nephropathy - refeeding syndrome - treatment of diabetic ketoacidosis with insulin

3.0 Management Strategies

Replace magnesium according to the table below (lower doses and close monitoring are required in patients with impaired renal function):

Severity	Treatment	Comments	Monitoring
MILD 0.5 to 0.7mmol/L*	Oral Replacement: 1) Magnesium aspartate 1000mg to 3000mg daily in divided doses (e.g. 1000mg daily to 1000mg TDS) 2) Magnesium (Elemental) 1 tab once to twice a day Administer with food	Formulation 1) Mag-sup® tablets contain 500mg of magnesium aspartate (37.4mg/1.55mmol of elemental magnesium) 2) Magnesium (Elemental) is made up oxide, phosphate pentahydrate and amino acid chelate salts (400mg/ 16.5mmol of elemental magnesium) Bioavailability of salts is variable - Aspartate = highest - Oxide = least Oxide salt associated with higher incidence diarrhoea	Magnesium Level (plasma/ serum) - Measure magnesium level (plasma/ serum) as required - Stool frequency (indication of excess dose)

MODERATE / SEVERE < 0.5mmol/L*	Intravenous Replacement 1) <i>Magnesium sulfate</i> 10mmol in 100mL of a compatible fluid administered over 60 minutes. 2) <i>Magnesium chloride</i> 10-20mmol in 250mL of a compatible fluid over 90-120minutes In High-Dependency Areas where cardiac monitoring is available, faster rates of infusion are allowed: • Magnesium sulfate 10mmol in 50mL of a compatible fluid administered over 30 minutes. • Repeat as required and titrate to effect and serum concentration. EMERGENCY SITUATIONS Can administer: • Magnesium sulfate 10mmol in 10mL of compatible fluid over 10 minutes² <i>Reference eTG²: 4 to 8 mmol magnesium can be given over 5 to 10 minutes - 10mmol over 10min is within this range</i>	Formulation 1) <i>Magnesium sulfate</i> • Preferred salt • Each 5mL contains 10mmol of magnesium ions 2) <i>Magnesium chloride</i> • Each 5mL contains 5mmol of magnesium ions Compatible fluids: Glucose 5%, glucose in sodium chloride solutions, Hartmann's, sodium chloride 0.9% • <i>Magnesium sulfate</i> diluted solutions are stable for 24 hours at 25°C • <i>Magnesium chloride</i> diluted solutions are to be used immediately	Magnesium Level (plasma/ serum) • Measure magnesium levels every 1 to 2 hours initially and then at least once daily whilst on IV therapy depending on severity of hypomagnesaemia and intensity of replacement. • Frequency of levels to be determined by the treating team • Please note: repletion of intracellular magnesium stores takes longer than normalisation of plasma/serum levels Toxicity Monitor for signs of hypermagnesaemia: blood pressure, heart rate, diminished reflexes, respiratory rate, oxygen saturation.
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*** The plasma/serum magnesium values above only serve as a guide; the presence or absence of symptoms should also be considered when determining whether oral, intravenous or no treatment is required**

4.0 Relative contraindications

If utilised for these indications specialist advice should be sought:

- Heart block
- Renal failure (creatinine clearance <20 mL/minute).

5.0 Precautions

- Impaired renal function increases the risk of hypermagnesaemia.
 - Dose reduction of at least 50% in those with creatine clearance <30ml/min

- The availability of an intravenous calcium salt as a reversal agent is recommended during magnesium sulfate administration in case of toxicity.
- Magnesium sulfate may precipitate an acute myasthenic crisis.

6.0 Adverse effects

- Diarrhoea (PO route)
 - often dose limiting side effect
 - most frequently with oxide salt, less frequent with aspartate salt
- Hypermagnesaemia
 - may manifest as blurred or double vision, CNS depression, nausea and vomiting, flushing, hypotension, loss of reflexes, muscle weakness, and muscle paralysis.
- Severe hypermagnesaemia (serum magnesium greater than 2.0-2.5mmol/l)
 - may cause cardiac arrest, cardiac arrhythmias, coma, respiratory depression, renal failure, and respiratory paralysis.

7.0 Compatibility

- Refer to the [Australian Injectable Drugs Handbook](#) for information.

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising policy and standards

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [Department of Health WA Mandatory Standards for Intravenous Potassium](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- [Australian Commission on Safety and Quality in Healthcare. Recommendation for terminology, abbreviations and symbols used in medicines documentation, December 2016](#)
- NSQHS Standards: 1,4,6,8

Standards, procedures, guidelines:

- [SCGH Nursing Practice Guideline No.51 Medication Management](#)
- [SCGH Nursing Practice Guideline No.4 Intravenous Therapy](#)

References

1. Yu ASL. Evaluation and treatment of hypomagnesemia [Internet]. Waltham, MA.: UpToDate; 2023 [cited 2023 July]. Available from: UpToDate.
2. Therapeutic Guidelines 2023 (online). West Melbourne. Therapeutic Guidelines Limited. eTG complete Therapeutic Guidelines Limited. March 2021. West Melbourne. Accessed online July 2023.
3. Harbord G. Specialised Drug Guideline – Magnesium. Fremantle Hospital and Health Service, 2013.
4. Symons K, Wong E (ed) Australian Injectable Drugs Handbook. 9th Ed. Melbourne: The Society of Hospital Pharmacists of Australia, 2023.
5. MIMS Online. St Leonards, NSW: UBM Medica; 2023 – Data Version July 2023.

6. Magnesium Complete Forte | Relaxation | Nutra-Life [Internet]. nutralife.com.au. [cited 2023 Aug 3]. Available from: <https://nutralife.com.au/product/magnesium-complete-forte/>
7. Blancquaert L, Vervaet C, Derave W. Predicting and Testing Bioavailability of Magnesium Supplements. *Nutrients*. 2019 Jul 20;11(7):1663.
8. StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022 Nov 4. Hypomagnesemia [cited 2023 August 16]. Available from [Hypomagnesemia - StatPearls - NCBI Bookshelf \(nih.gov\)](https://www.ncbi.nlm.nih.gov/books/NBK537031/)

Authorisation & Version Control

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission

This document can be made available in alternative formats on request for a person with a disability.



BRIEF SUMMARY

DRUG	INDICATION	PRESENTATION	PREPARATION	DOSAGE	ADMINISTRATION	COMMENT
Magnesium aspartate tablets	Mild hypomagnesaemia (0.5-0.7mmol/L)	500mg tablet contains 37.5mg/1.55mmol of elemental magnesium		1000mg-3000mg (i.e. 2-6 tabs) in daily divided doses	Take with/soon after food	Diarrhoea common dose limiting side effect - oxide salt highest incidence
Magnesium (elemental) Forte capsules		Capsule contains 400mg/16.5mmol of elemental magnesium		1 capsule once to twice a day	Tablets can be crushed Capsules can be open and content crushed	Highly variable bioavailability different salts -aspartate highest bioavailability
Magnesium sulfate injection	Moderate-to-severe hypomagnesaemia (preferred salt)	Ampoule containing a clear, colourless solution of 10mmol magnesium in 5mL	10mmol in 100mL of sodium chloride 0.9%/glucose 5% High dependency areas with cardiac monitoring 10mmol in 50mL of sodium chloride 0.9%/glucose 5%	Dose dependant on plasma/ serum level Regular monitoring and subsequent dose adjustment required Approx dose/day Level <0.4mmol/L • 20-40mmol Level 0.4 to 0.6mmol/L • 10-30mmol Level >0.6mmol/L • 10-20mmol	Infuse over at least 60mins High dependency with cardiac monitoring Infuse of at least 30mins Emergency Situations Over 10mins	Regular monitoring of magnesium levels required Ensure adequate mixing by invert and shake infusion bag several times after preparing infusion
Magnesium chloride injection	Moderate-to-severe hypomagnesaemia	Ampoule containing a clear, colourless solution of 5mmol magnesium in 5mL	10-20mmol in 250mL of sodium chloride 0.9%/glucose 5%		10mmol infuse over 90mins 20mmol infuse over 120mins	Chloride salt reserved for sulfate salt intolerance